

# SMART INDIA HACKATHON 2026

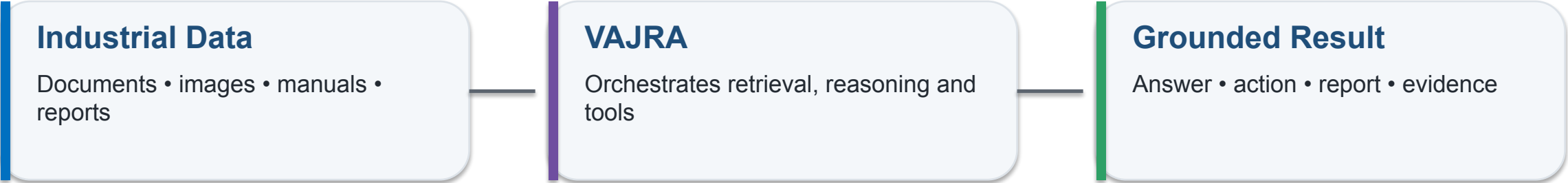


## TITLE PAGE

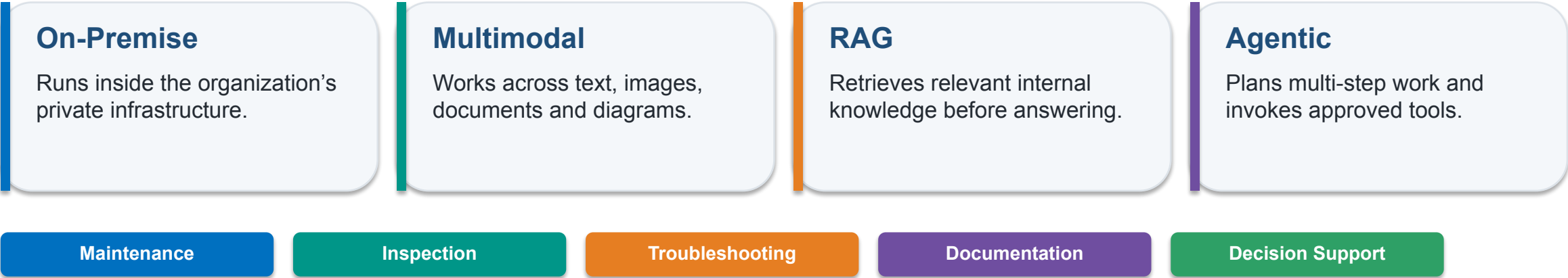
- **Problem Statement ID** – SIH26117
- **Problem Statement Title-** Sovereign On-premise Agentic AI Workbench using open-weight Multimodal LLMs for Confidential Industrial work
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-** A36
- **Team Name** - IGRIS



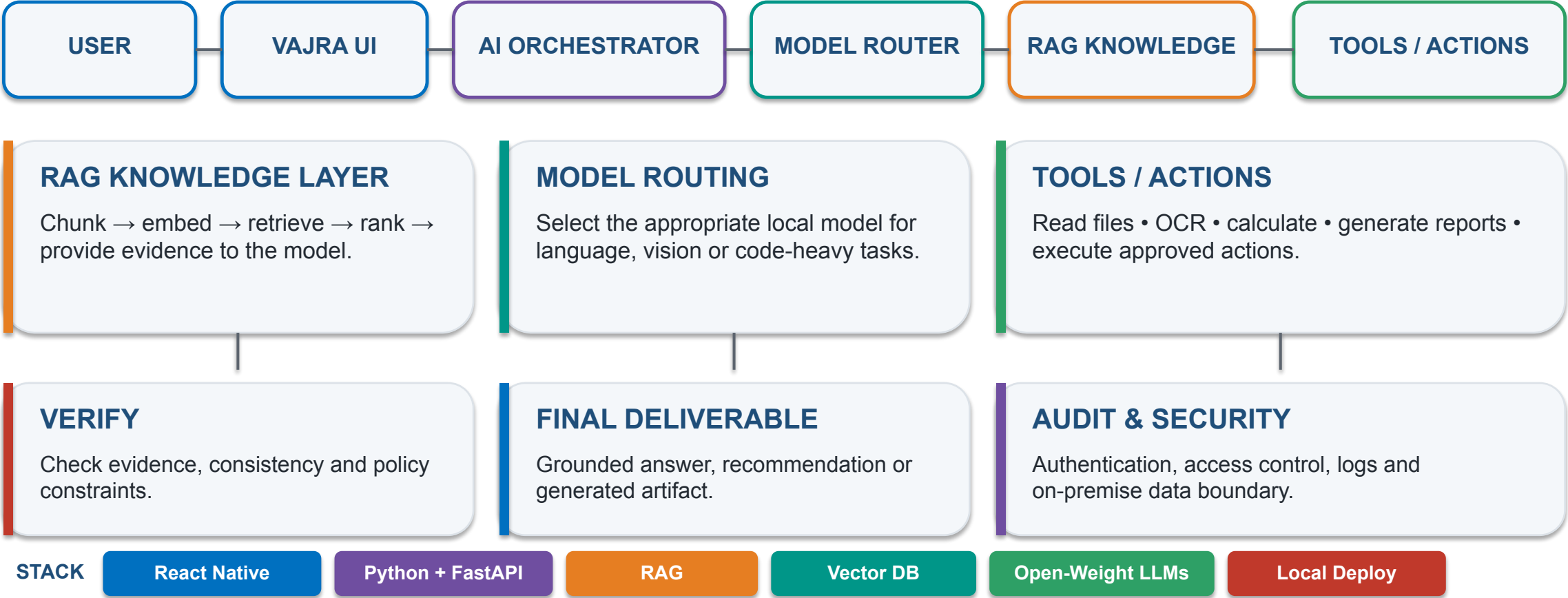
## WHAT VAJRA DOES



## CORE APPROACH



## END-TO-END TECHNICAL WORKFLOW





# FEASIBILITY AND VIABILITY



## CAN VAJRA BE DEPLOYED IN THE REAL WORLD?

### FEASIBILITY

**Local inference**  
Open-weight models can run within controlled infrastructure.

**Existing software stack**  
Python/FastAPI + vector retrieval supports rapid integration.

**Incremental deployment**  
Start with documents and RAG, then add tools and multimodal workflows.

### VIABILITY

**Data sovereignty**  
Sensitive industrial information stays inside the organization.

**Lower recurring API cost**  
Reduces dependence on per-call external AI services.

**Department-ready**  
Knowledge and tools can be expanded use-case by use-case.

### KEY CONSTRAINTS

**Compute**  
Model size and local inference speed depend on available GPU/CPU.

**Integration**  
Industrial databases and repositories need controlled connectors.

**Accuracy & governance**  
Retrieval quality, permissions and verification must be tested.

### DEPLOYMENT PATH



## IMPACT, NOVELTY & SCALABILITY

### FASTER OPERATIONS

Target: reduce time spent searching manuals, reports and internal knowledge.

### LESS MANUAL EFFORT

Target: automate repetitive retrieval, analysis and documentation steps.

### LOWER EXTERNAL DEPENDENCY

Target: keep sensitive workloads and recurring inference inside the organization.

## NOVELTY / USP

### Why VAJRA is different

- Sovereign by design
- Agentic + RAG + multimodal
- Local model routing
- Tool execution with verification
- Auditable workflow

## BUSINESS VALUE

### Where value appears

- Maintenance knowledge access
- Inspection assistance
- Troubleshooting support
- Automated documentation
- Decision support from internal evidence

## SCALABILITY

### Expansion model

One use case → multiple departments → multiple plants, while retaining the same security and orchestration layer.

One Use Case

Multiple Departments

Multiple Plants

Scale

## RESEARCH & REFERENCES

### ACADEMIC / TECHNICAL REFERENCES

#### RAG

Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks” (2020)

#### Agentic workflow

Yao et al., “ReAct: Synergizing Reasoning and Acting in Language Models” (2022)

#### Multimodal LLMs

Liu et al., “Visual Instruction Tuning” / LLaVA (2023)

### DEVELOPMENT / PRIOR-ART SOURCES

#### Google Colab

Prototype environment used for RAG/model experimentation.

#### Kaggle

Open datasets used for experimentation and testing.

#### Open-weight models

Model cards and official documentation for local deployment and evaluation.

#### Industrial AI

Enterprise knowledge-retrieval and on-premise AI patterns informing the architecture.

### RESOURCES & QUICK LINKS

[colab.research.google.com](https://colab.research.google.com)   [kaggle.com](https://kaggle.com)   [arxiv.org](https://arxiv.org)

*Note: Add your final project repository, model cards, and exact paper URLs before submission.*